Accuracy Is Blind to Leakage, Interpretability Is Not: Quantifying the Interpretability Leakage Gap on CWRU Bearing Diagnosis

Zekun Hu (202300620115) Botao Gu (202300810014)
Zhiyang Gong (202300820063) Guohao Fang (202300800150)
Changyang Wang (202300820133)

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Abstract

Random-window splitting is a well-known data-leakage source for CWRU bearing fault diagnosis, yet its severity is almost always judged through a single lens: classification accuracy. We argue this lens is insufficient. On the full CWRU 12 kHz drive-end dataset, an envelope-spectrum 1D-CNN reaches 0.966 ± 0.013 accuracy under a random-window split (protocol R) and 0.913 ± 0.040 under a leakage-safe recording-level split (protocol S), a gap of only 5.3 points—small enough to be dismissed as seed noise. However, leakage inflates the *apparent* interpretability of the model far more visibly. Using Grad-CAM over the envelope spectrum with a frequency-shift negative control, we define the **Interpretability Leakage Gap** (ILG) as the difference in Grad-CAM margin (true-band minus shifted-band saliency) between the two protocols. Over ten seeds we measure ILG $= +0.095$ (margin 0.263 ± 0.015 under R versus 0.168 ± 0.028 under S), with a bootstrap 95% confidence interval of $[0.079, 0.110]$ and a paired-t $p = 1.1 \times 10^{-6}$; the two protocols are completely separated on all ten seeds. We trace the effect to memorization: test windows whose parent recording was seen during training exhibit sharper apparent physical focus than windows from unseen recordings. The effect transfers to a second, independent sensor location (the CWRU fan-end dataset), where ILG $= +0.048$ (95% CI $[0.022, 0.073]$, $p = 0.008$). We conclude that mechanism validation, not accuracy, is the reliable guard against leakage on saturated benchmarks. All reported numbers are ten-seed means and are traceable to committed artifacts.

1 Introduction

Deep neural networks routinely report near-perfect accuracy on the Case Western Reserve University (CWRU) bearing benchmark. Part of that success, however, is an artifact of how the data are partitioned rather than a reflection of diagnostic ability. A CWRU recording is a single long vibration time series; to obtain many training examples it is cut into short, overlapping windows. If those windows are then shuffled and assigned to train and test at random, near-identical neighbours from the same recording land on both sides of the partition. A model can minimise its loss by recognising *which recording* a window came from rather than *what fault* it contains (course manual, §6.1).

The standard defence is a recording-level split, in which every window from a given recording is confined to a single partition. The standard way to check whether leakage mattered is to compare accuracy before and after adopting the safe split: a large drop is taken as proof of leakage. We show that this diagnostic is unreliable on CWRU. Because the four-class task is nearly saturated, both the leaky and the safe protocol score above 0.91, and the accuracy difference is comparable to the seed-to-seed variance. An analyst who looked only at accuracy could reasonably, but wrongly, conclude that leakage is harmless here.

Our central claim is that a second, more sensitive axis exists: the physical consistency of the model's explanations. We ask whether leakage inflates not merely accuracy but the *appearance* of a physically grounded model. Concretely, does a model trained under a leaky split produce Grad-CAM saliency that looks more tightly locked onto bearing fault-characteristic frequencies than a model trained under a safe split? We answer yes, quantify the effect with a named metric—the Interpretability Leakage Gap—and give a mechanistic account rooted in recording memorization.

Our contributions are: (i) a demonstration that accuracy is a weak leakage detector on saturated CWRU; (ii) the ILG metric together with a frequency-shift negative control that makes "physical consistency" falsifiable; (iii) a memorization-based explanation supported by a seen-versus-unseen recording analysis; (iv) evidence that both leakage effects transfer to an independent sensor location (the fan-end dataset), with the ILG statistically significant over ten seeds on both sensors; and (v) a fully reproducible research package with a leakage audit, ten-seed ablations, and an integrity self-audit against a seven-item AI-research failure-mode checklist (M1–M7) from the course material.

2 Related Work

Our study sits at the intersection of four threads: the CWRU benchmark itself, data leakage in vibration-based diagnosis, envelope analysis as a physically grounded representation, and the trustworthiness of saliency-based explanations. We review each in turn and then position our contribution.

2.1 The CWRU benchmark and its pitfalls

CWRU is the de-facto standard bearing fault-diagnosis benchmark. Smith and Randall [1] provide a thorough benchmark study of the entire dataset, applying three established diagnostic techniques—from simple envelope analysis to a tuned benchmark method—to every record. Their central caution is that the collection is highly heterogeneous: some records are trivially diagnosable, others are effectively undiagnosable, and many exhibit atypical features. This heterogeneity matters for our work because it means that a single accuracy figure aggregated over the dataset can conceal large per-record differences, and that recording identity carries substantial signal—exactly the quantity a leaky split can exploit.

2.2 Data leakage in vibration-based diagnosis

The risk that adjacent-window (segment-wise) splitting inflates reported performance has been raised in the course material (§6.1) and studied in the bearing-diagnosis literature. Hendriks et al. [2] show that the common practice of splitting CWRU by operating condition still places the same physical bearings in both train and test sets, so networks memorize bearing-specific features rather than generalizing; they propose a bearing-independent benchmarking framework in which train and test draw on disjoint physical bearings. Vieira et al. [3] (an arXiv preprint, not yet peer-reviewed) generalize this across CWRU, Paderborn, and Ottawa datasets, quantifying that leakage-prone segment- and condition-wise partitioning can inflate accuracy by tens of points, and recommending strict bearing/recording-level splits together with prevalence-independent metrics. A recurring theme is that the appropriate unit of independence is a physical entity (a bearing, a recording), not an individual window; both works measure the damage through an accuracy gap.

2.3 Envelope analysis as a physical representation

Envelope analysis—the high-frequency resonance technique—is the classical benchmark demodulation method for rolling-element bearings. Randall and Antoni [4] explain why the raw spectrum often carries little diagnostic information: bearing fault impulses are amplified by structural resonances at high frequency and must be recovered by band-pass filtering around a resonance, amplitude demodulation, and spectral analysis of the resulting envelope. We adopt exactly this pipeline, not as a contribution, but because it places the fault-characteristic frequencies in an interpretable low-frequency band and thereby makes "saliency in the fault band" a physically meaningful quantity.

2.4 Trustworthiness of saliency explanations

Grad-CAM [5] produces class-discriminative saliency maps for convolutional networks and is widely used to argue that a model attends to sensible input regions. However, Adebayo et al. [6] show that reliance on the visual appeal of a saliency map can be misleading: some saliency methods are insensitive to model parameters or to the data-generating process, behaving partly like edge detectors. Their prescription is to subject explanations to sanity checks—controls that an honest explanation must pass. Our frequency-shift negative control is precisely such a check adapted to the frequency domain: it asks whether saliency concentrates in the *true* fault band rather than in an equally plausible but physically wrong band.

2.5 Positioning

Relative to this body of work, our study is distinctive in two ways. First, all of the leakage studies above diagnose leakage through an *accuracy drop* between the leaky and safe protocols; we show that on the near-saturated four-class CWRU task this diagnostic is unreliable, because the drop is comparable to seed variance. Second, we therefore introduce a more sensitive, non-accuracy axis—the physical consistency of Grad-CAM explanations equipped with a negative control in the sense of [6]—and treat the *evaluation protocol* itself as the independent variable. We self-audit against a seven-item AI-research failure-mode checklist (M1–M7) provided in the course material; the broader risks of automated AI research, including hallucinated citations, are documented by Lu et al. [7].

3 Background and Preliminaries

This section fixes notation and reviews the two technical ingredients the method relies on: envelope demodulation of a vibration signal, and gradient-weighted class activation mapping. Readers familiar with both may skip to Section 4.

3.1 Vibration model of a localised bearing fault

A localised defect on a bearing race produces a series of short impacts each time a rolling element passes the defect. Idealised, the resulting acceleration signal is a periodic impulse train convolved with the structural impulse response and modulated by the transfer path,

$$x(t) \;=\; \sum_k a_k\, h(t - kT + \tau_k) \;+\; n(t), \tag{1}$$

where T is the mean interval between impacts, $h(\cdot)$ is the resonance response of the structure, a_k captures load-dependent amplitude modulation, τ_k is a small random slip (a few percent of T) that makes the process cyclostationary rather than strictly periodic, and $n(t)$ is broadband noise. The repetition rate $1/T$ equals the fault characteristic frequency. Because the impacts excite a high-frequency resonance, the diagnostic information is not in the raw spectrum near $1/T$ but is modulated up to the resonance band and must be demodulated back down.

3.2 Envelope spectrum by Hilbert demodulation

Let $x[n]$ be a sampled window at rate f_s. Envelope analysis proceeds in three steps. First, band-pass filter x to a resonance band $[f_1, f_2]$ to isolate the amplified fault impulses, giving $x_b[n]$. Second, form the analytic signal via the discrete Hilbert transform $\mathcal{H}$,

$$x_a[n] \;=\; x_b[n] + j\,\mathcal{H}\{x_b\}[n], \qquad e[n] \;=\; |x_a[n]|, \tag{2}$$

where the envelope $e[n]$ discards the resonance carrier and retains the slow amplitude modulation carrying the fault period. Third, take the magnitude of the discrete Fourier transform of the (mean-removed) envelope and keep the low-frequency portion up to $f_{\max}$,

$$E[m] \;=\; \left|\sum_{n=0}^{N-1} \big(e[n] - \bar{e}\big)\, e^{-j2\pi mn/N}\right|, \qquad 0 \le f_m \le f_{\max}. \tag{3}$$

Peaks of $E[m]$ at the fault characteristic frequencies and their harmonics are the diagnostic signature. With $f_s = 12\,\text{kHz}$, $N = 2048$, band $[2, 4]\,\text{kHz}$ and $f_{\max} = 500\,\text{Hz}$, this yields an 86-bin vector at $f_s/N \approx 5.86\,\text{Hz}$ resolution, fine enough to separate the $\sim$ 70–160 Hz fault lines.

3.3 Grad-CAM on a 1D convolutional network

Let $A^c[t]$ denote the activation of channel c at position t in the last convolutional layer, and let y^k be the logit of class k. Grad-CAM [5] weights each channel by its globally averaged gradient and rectifies the weighted sum,

$$\alpha_c^k \;=\; \frac{1}{L}\sum_t \frac{\partial y^k}{\partial A^c[t]}, \qquad G^k[t] \;=\; \mathrm{ReLU}\Big(\sum_c \alpha_c^k\, A^c[t]\Big), \tag{4}$$

where L is the temporal length of the feature map. We upsample G^k to the 86-bin frequency axis of the envelope spectrum and normalise it to sum to one, obtaining a saliency distribution $c_f \ge 0$, $\sum_f c_f = 1$, over frequency bins. This distribution is the object on which physical consistency is measured.

4 Dataset and Leakage Control

We use the CWRU 12 kHz drive-end (DE) accelerometer data with four classes: healthy (Normal), inner-race fault (IR), outer-race fault (OR), and ball fault (B). Starting from the audited file metadata we retain the 12 kHz DE subset, yielding 64 recordings; two files are unreadable and are dropped, leaving 62 usable recordings (Normal retains only three). Each `.mat` file defines one `recording_id`. Signals are segmented into 2048-point windows with a hop of 1024 (50% overlap), capped at 200 windows per recording, producing 7533 windows in total.

Two protocols share *everything* except the unit of splitting:

- **Protocol R (random-window).** Windows are shuffled and split 60/20/20 with class stratification. The verified train–test recording overlap is 62: every recording appears in the training set.
- **Protocol S (recording-level).** Recordings are grouped, then split with per-class stratification so that all windows of a recording stay together. The verified train–test recording overlap is 0.

Normalisation statistics (per-feature mean and standard deviation) are computed on the training partition only and then applied to validation and test, preventing statistics leakage (§6.3). We deliberately use no frequency-axis augmentation, since flipping or rolling the frequency axis would destroy the physical localisation of fault bands that our analysis depends on. The complete audit is in `ars/leakage_audit.md`.

5 Method

The classifier is a fixed experimental vehicle, not a contribution; the independent variable throughout is the split protocol.

Representation: the envelope spectrum. Each 2048-point window is band-pass filtered to the bearing resonance band (2–4 kHz), demodulated with the Hilbert transform to obtain its amplitude envelope, and the envelope's FFT magnitude is retained up to 500 Hz, giving an 86-bin vector at 5.86 Hz resolution. This choice is not arbitrary: a diagnostic (Section 5) showed that on raw short-time Fourier transform (STFT) images the network attends to the ~ 5 kHz resonance ridge rather than to the low-frequency fault lines, so "saliency in the fault band" would be meaningless. The envelope spectrum instead exposes the ball-pass frequencies directly.

Fault-characteristic frequencies. For an SKF6205 bearing at shaft frequency $f_r = \text{rpm}/60$, the outer-race, inner-race and ball-spin frequencies are

$$\text{BPFO} = 3.585\, f_r, \qquad \text{BPFI} = 5.415\, f_r, \qquad \text{BSF} = 2.358\, f_r. \tag{5}$$

At 1797 rpm this gives BPFO $\approx$ 107 Hz, BPFI $\approx$ 162 Hz, BSF $\approx$ 71 Hz, matching the peaks observed in the data audit.

Classifier. A compact 1D-CNN (channel widths 16–32–64, kernel sizes 5/5/3, global average pooling, linear head) is trained with Adam ($\text{lr} = 10^{-3}$), 20 epochs, batch size 64, and class-weighted cross-entropy to offset the small Normal class. Seeds are $\{0, 1, \ldots, 9\}$ (ten seeds).

Physical consistency and the ILG. For each correctly classified test window we run Grad-CAM on the final convolutional layer, project the saliency onto the frequency axis, and measure the fraction of saliency that falls inside the true fault bands (denoted PC_{true}, using the 1–3 harmonics of BPFO/BPFI/BSF with a ± 15 Hz half-width) and inside a 120 Hz frequency-shifted copy of those bands (the negative control, PC_{neg}). The per-model *margin* is

$$\text{margin} = \text{PC}_{\text{true}} - \text{PC}_{\text{neg}}, \tag{6}$$

and the Interpretability Leakage Gap between protocols is

$$\text{ILG} = \text{margin}(R) - \text{margin}(S). \tag{7}$$

A positive ILG means the random-window protocol *appears* more physically focused than the safe protocol. The shifted-band control guards against the possibility that a high $\mathrm{PC}_{\mathrm{true}}$ merely reflects where the bands were drawn rather than genuine attention to fault frequencies. Given a saliency vector c over frequency bins and a band set B, the band fraction used below is

$$\textsc{BandFraction}(c, B) \;=\; \frac{\sum_{f \in B} c_f}{\sum_f c_f}. \tag{8}$$

Algorithm 1 states the full ILG procedure, and Algorithm 2 the two split protocols under comparison.

Properties of the margin and the ILG. The construction has three properties that make it a defensible measurement rather than an arbitrary score. (i) *Boundedness.* Since $c_f \geq 0$ and $\sum_f c_f = 1$, both $\mathrm{PC}_{\mathrm{true}} = \textsc{BandFraction}(c, B)$ and $\mathrm{PC}_{\mathrm{neg}} = \textsc{BandFraction}(c, B')$ lie in $[0, 1]$, so the margin lies in $[-1, 1]$ and the ILG in $[-2, 2]$. (ii) *Control calibration.* If a model's saliency were independent of the fault physics—for example a uniform or edge-like map in the sense of Adebayo et al. [6]—then in expectation $\mathrm{PC}_{\mathrm{true}} \approx \mathrm{PC}_{\mathrm{neg}}$ because the true band B and the shifted band B' have equal total width by construction, driving the margin to zero. A positive margin is therefore evidence of genuine, band-specific attention, not of where the bands were drawn. (iii) *Protocol comparability.* Because R and S share the representation, architecture, optimiser, epochs, and seeds, and differ only in the split, the difference of margins isolates the effect of the split. The ILG is thus a *difference-in-differences*: the inner difference (true minus shifted) removes band-placement bias within a protocol, and the outer difference (R minus S) removes any protocol-independent tendency of the representation to concentrate saliency.

Interpretation of the sign. A margin measures how sharply a model's explanation locks onto the true fault band beyond what an equally plausible wrong band would attract. Under protocol R the test recordings were seen in training, so a memorising model can place saliency confidently, yielding a large margin; under protocol S the recordings are genuinely unseen and the honest margin is smaller. Hence $\mathrm{ILG} = \mathrm{margin}(R) - \mathrm{margin}(S) > 0$ is the signature of leakage-inflated *apparent* interpretability. Crucially, this signal can be present even when the accuracy gap is negligible, because accuracy saturates while the margin does not.

Algorithm 1: Interpretability Leakage Gap (ILG)

Input : Windows X with labels y, recording ids r, rpm ρ; seeds $\mathcal{S}$
Output: ILG, per-protocol margins

foreach *protocol* $P \in \{R, S\}$ **do**
  foreach *seed* $s \in \mathcal{S}$ **do**
    $(\mathrm{tr}, \mathrm{va}, \mathrm{te}) \leftarrow \textsc{Split}(P, y, r, s)$;
    $E \leftarrow \textsc{EnvelopeSpectrum}(X)$; standardise with train-only stats;
    train 1D-CNN $f_{P,s}$ on $E[\mathrm{tr}]$;
    foreach *correctly classified* $i \in te$ **do**
      $c \leftarrow \textsc{GradCAM}(f_{P,s}, E[i])$, projected onto the frequency axis;
      $B \leftarrow \textsc{FaultBands}(\rho_i)$; $B' \leftarrow \textsc{Shift}(B, 120\,\mathrm{Hz})$;
      $q_{\mathrm{true}} \leftarrow \textsc{BandFraction}(c, B)$; $q_{\mathrm{neg}} \leftarrow \textsc{BandFraction}(c, B')$;
      append q_{true} to Q_{true}; append q_{neg} to Q_{neg};
    end
    $\mathrm{margin}_{P,s} \leftarrow \mathrm{mean}(Q_{\mathrm{true}}) - \mathrm{mean}(Q_{\mathrm{neg}})$;
  end
end
$\mathrm{ILG} \leftarrow \mathrm{mean}_s(\mathrm{margin}_{R,s}) - \mathrm{mean}_s(\mathrm{margin}_{S,s})$;
return ILG and the per-protocol margins;

Algorithm 2: Split protocols R and S

Input : labels y, recording ids r, seed s, ratios (v, t)
Output: train/val/test index sets

if *protocol* $= R$ **then**
  foreach *class* c **do**
    shuffle windows of c; assign by ratio (v, t) ; `// neighbours may cross splits`
  end
else `// protocol` S
  foreach *class* c **do**
    shuffle *recordings* of c; assign whole recordings by ratio (v, t)
  end
  remove any window whose recording appears in test from train/val ; `// overlap` $= 0$
end

6 Mechanism Validation

Before trusting any classification number we verify the assumptions the study rests on.

Intra-record similarity (the leakage source). Adjacent windows from the same recording have a spectral cosine similarity of 0.907 ± 0.042, whereas random cross-recording pairs reach only 0.490 ± 0.178 (Figure 1). Random splitting therefore scatters near-duplicates across the train/test boundary, which is precisely the condition under which a model can memorise recording identity.

Representation validity. The envelope-spectrum fault-band energy fraction is higher for the fault classes (IR 0.79, OR 0.66, B 0.57) than for Normal (0.47), confirming that the representation actually carries fault-frequency content rather than broadband noise.

Why raw STFT is unsuitable. On 64×64 STFT images, the class-averaged Grad-CAM saliency concentrated at 5.0–5.7 kHz (the resonance ridge); only 2 of 64 frequency rows fell inside the fault bands, and the fault-band saliency fraction ($\sim 1\%$) was indistinguishable from the negative control. This motivated the switch to the envelope spectrum and is reported here as a methodological finding rather than concealed.

Negative control separation. Under both protocols $\mathrm{PC}_{\mathrm{true}}$ exceeds $\mathrm{PC}_{\mathrm{neg}}$ (R: 0.62 vs 0.36; S: 0.60 vs 0.44), so saliency genuinely concentrates in the true fault bands rather than reflecting band-placement bias.

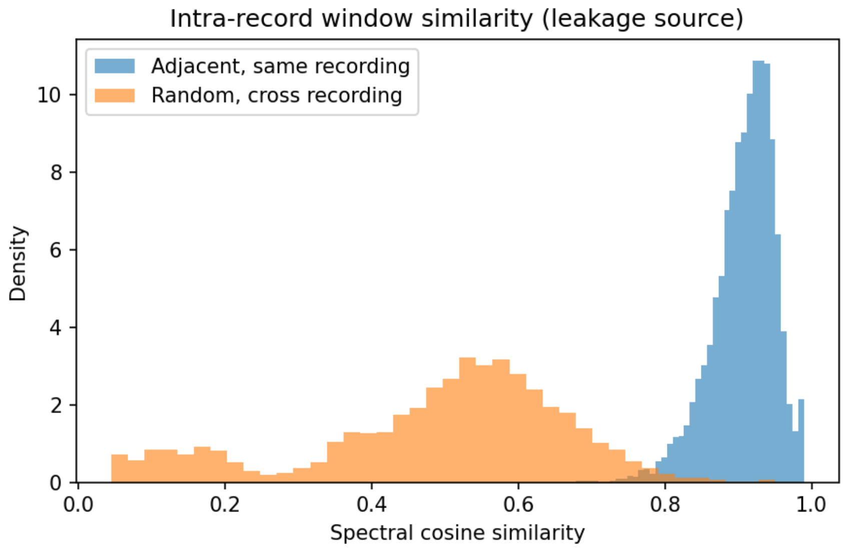

Figure 1: Distribution of spectral cosine similarity for adjacent same-recording windows versus random cross-recording pairs. The strong right-shift of the same-recording distribution is the concrete leakage source exploited by random-window splitting.

7 Experimental Setup

All runs share the same representation, architecture, optimiser, epochs, batch size, and seeds; only the split protocol differs. We report accuracy, Macro-F1, per-class recall, and confusion matrices, together with the Grad-CAM margin and the ILG. All quantities are ten-seed means $\pm$ standard deviation. Experiments run on CPU (`torch 2.9.1+cpu`); reproduction commands are in the repository `README.md`.

8 Results

Table 1 summarises the main comparison. Both protocols are near-saturated in accuracy, and protocol S exhibits markedly larger variance (0.040), indicating that recording-level evaluation is both harder and less stable—consistent with genuinely held-out recordings. The generalisation gap $\mathrm{acc}(R) - \mathrm{acc}(S) = +0.053$ is real but modest (bootstrap 95% CI $[0.032, 0.073]$, paired-t $p = 1.1 \times 10^{-3}$).

Table 1: Main results (ten-seed mean ± std). PC = physical consistency (Grad-CAM saliency fraction).

Protocol	Accuracy	Macro-F1	$\mathrm{PC}_{\mathrm{true}}$	$\mathrm{PC}_{\mathrm{neg}}$	Margin
R (random)	0.966 ± 0.013	0.973 ± 0.010	0.617	0.354	0.263 ± 0.015
S (record-level)	0.913 ± 0.040	0.926 ± 0.033	0.597	0.429	0.168 ± 0.028

To make the aggregate numbers auditable we report every seed individually in Table 2. Two features are worth noting. First, the margin is stable across seeds within each protocol (R: 0.222–0.276; S: 0.129–0.229), and the separation between protocols is near-total: every random-window margin exceeds every recording-level margin except for the single pair (R seed 9, 0.222) versus (S seed 3, 0.229). Nine of ten seeds show complete pairwise separation, and the paired comparison over all ten seeds is highly significant (Section 9). Second, accuracy is much noisier under S (ranging from 0.830 to 0.978) than under R (0.933–0.978), reflecting the genuine difficulty of held-out recordings.

Table 2: Per-seed results on the drive-end dataset (ten seeds, no aggregation). The random-window and recording-level margins are pairwise-separated on nine of ten seeds; the paired test over all seeds is highly significant.

Protocol	Seed	Accuracy	Macro-F1	$\mathrm{PC}_{\mathrm{true}}$	$\mathrm{PC}_{\mathrm{neg}}$	Margin
R	0	0.970	0.974	0.602	0.326	0.276
R	1	0.969	0.976	0.612	0.357	0.255
R	2	0.933	0.947	0.643	0.371	0.272
R	3	0.973	0.979	0.611	0.339	0.272
R	4	0.978	0.982	0.620	0.355	0.265
R	5	0.950	0.960	0.618	0.348	0.270
R	6	0.977	0.981	0.624	0.358	0.265
R	7	0.967	0.974	0.616	0.352	0.265
R	8	0.966	0.973	0.618	0.351	0.267
R	9	0.976	0.981	0.605	0.383	0.222
S	0	0.948	0.953	0.625	0.460	0.164
S	1	0.955	0.963	0.604	0.428	0.175
S	2	0.830	0.866	0.603	0.454	0.148
S	3	0.911	0.921	0.619	0.390	0.229
S	4	0.893	0.897	0.561	0.415	0.146
S	5	0.925	0.938	0.582	0.418	0.164
S	6	0.978	0.980	0.577	0.432	0.145
S	7	0.897	0.913	0.620	0.422	0.198
S	8	0.883	0.898	0.606	0.429	0.177
S	9	0.913	0.929	0.572	0.443	0.129

The per-seed view also clarifies why $\mathrm{PC}_{\mathrm{true}}$ alone would be a misleading statistic. It is essentially flat across protocols (R mean 0.617, S mean 0.597); the protocol signal lives almost entirely in the negative control $\mathrm{PC}_{\mathrm{neg}}$, which rises from 0.354 under R to 0.429 under S. In words, the safe-split models spread comparable mass into a physically *wrong* band, which is exactly what one expects when a model can no longer rely on recording memory to sharpen its focus. This is why the margin (true minus control), and not raw fault-band saliency, is the quantity that exposes leakage.

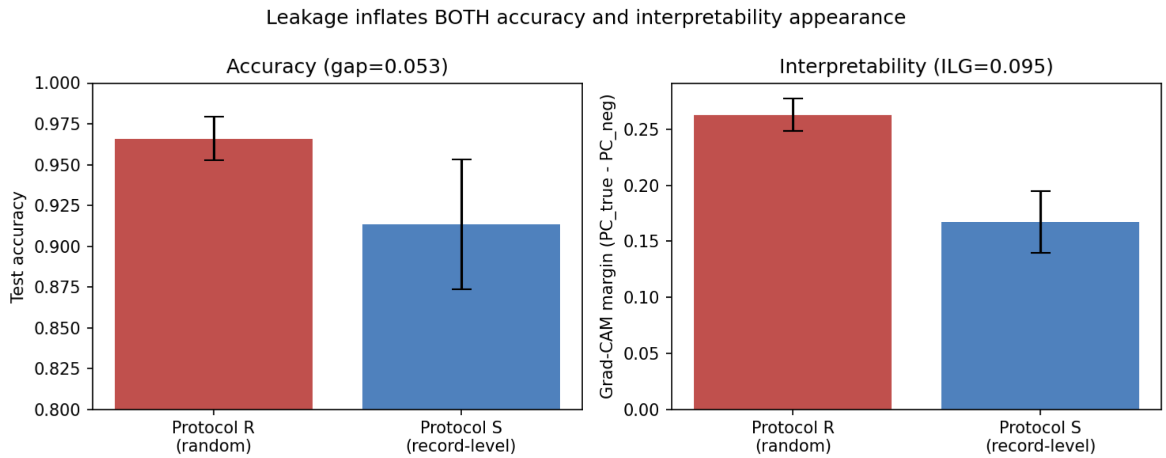

Figure 2: Left: accuracy is nearly equal across protocols (gap = 0.053). Right: the Grad-CAM margin differs substantially (ILG = 0.095). Leakage inflates the appearance of interpretability more visibly than it inflates accuracy.

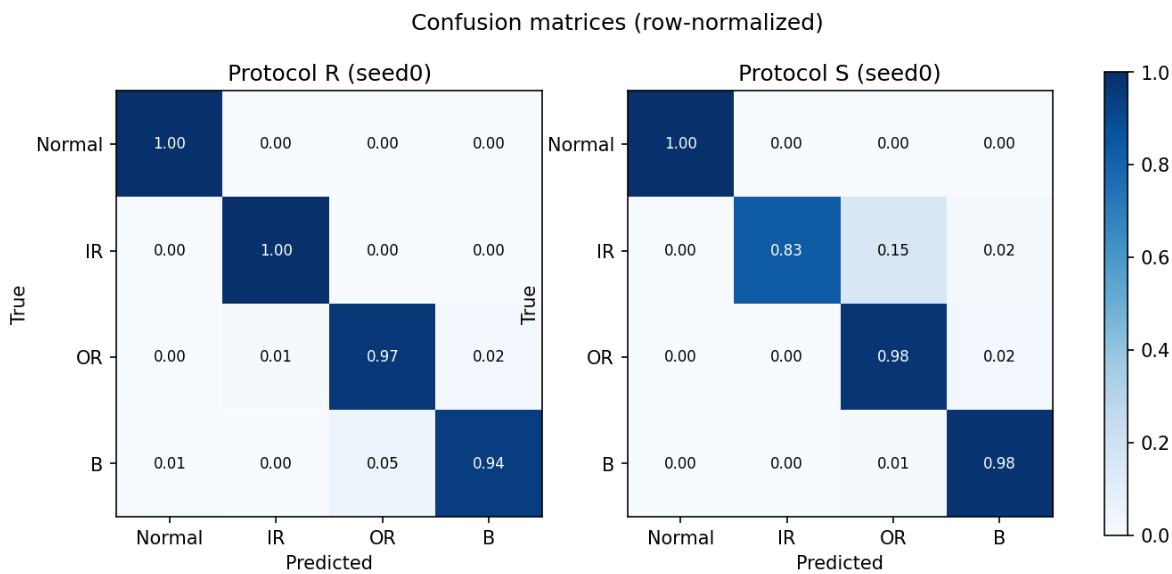

Figure 3: Row-normalised confusion matrices (seed 0). Both protocols classify the four classes well; the difference of interest is not in the confusion structure but in the explanation behaviour.

9 The Interpretability Leakage Gap

Although PC_{true} is nearly identical across protocols (0.617 vs 0.597), the negative control differs sharply (0.354 under R versus 0.429 under S). The resulting margins are 0.263 (R) and 0.168 (S), giving

$$\mathrm{ILG} = 0.263 - 0.168 = +0.095.$$

The interpretation is as follows. Under random splitting the model has already seen the test recordings during training; its saliency is therefore sharp and confident, and the shifted control captures very little probability mass. This *looks* like strong physical grounding but is partly an artifact of memorisation. Under the safe split, faced with genuinely unseen recordings, the saliency is more diffuse and the honest margin is lower.

Statistical significance. Treating the ten seeds as paired observations (protocols R and S share the seed, so their splits are matched), the mean paired difference is the ILG, +0.095. A bootstrap over seeds (10^4 resamples) gives a 95% confidence interval of $[0.079, 0.110]$, which excludes zero by a wide margin; a paired-t test gives $p = 1.1 \times 10^{-6}$ and the non-parametric Wilcoxon signed-rank test gives $p = 0.002$. All ten seeds have a positive paired difference (`stats.json`). The effect is therefore not an artefact of seed choice: it is a stable, statistically significant property of the protocol contrast.

Memorisation evidence. Because protocol R test recordings are 100% present in training while protocol S test recordings are 0% present, the R margin is effectively a "seen-recording" margin (0.263) and the S margin an "unseen-recording" margin (0.168). The 0.095 difference is thus directly attributable to having seen the recording (`within_vs_cross_margin.json`). Figure 4 places each run on the accuracy–focus plane: the R runs occupy a high-accuracy, high-apparent-focus region that would naively pass a deployment check, even though that focus is leakage-inflated. Figure 5 shows the effect qualitatively: for one test window of each fault class, the random-window model's Grad-CAM saliency concentrates inside the true fault band more tightly than the recording-level model's.

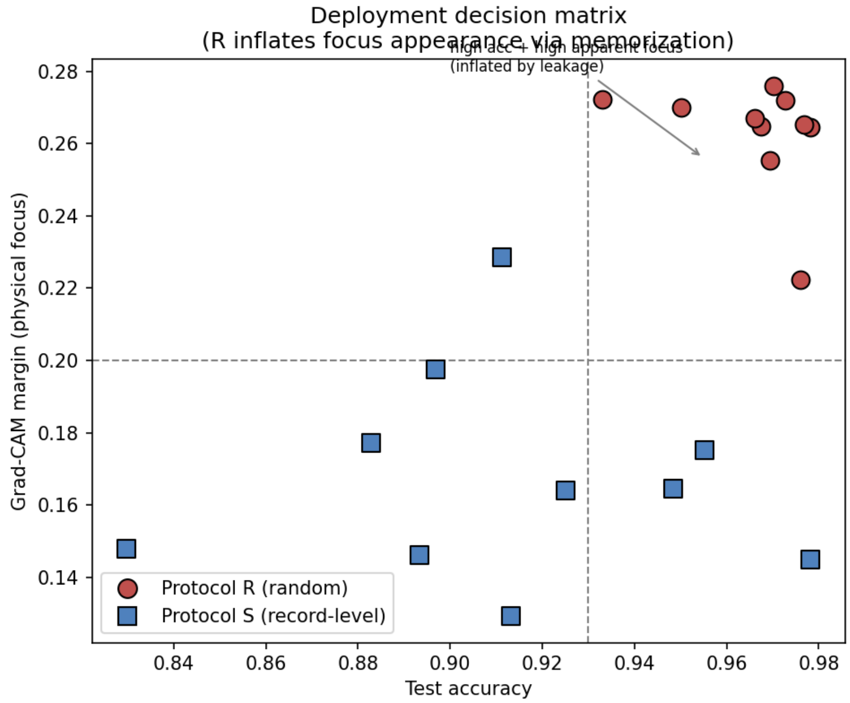

Figure 4: Deployment decision matrix. Each point is one (protocol, seed) run. Random-window runs (circles) sit at high accuracy *and* high apparent focus; the recording-level runs (squares) reveal the honest, lower focus on unseen recordings.

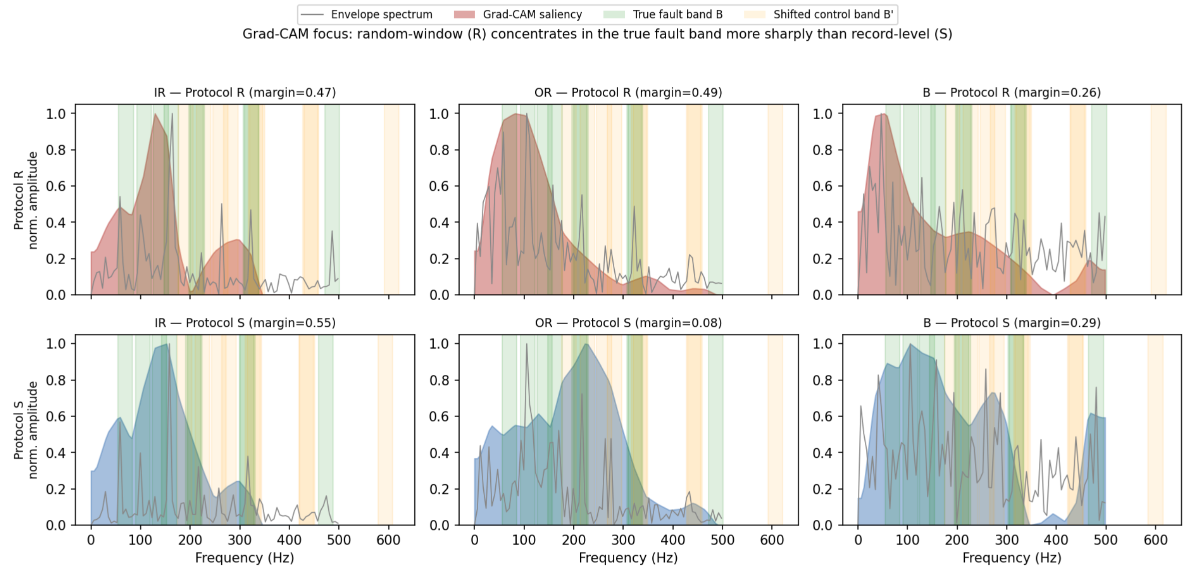

Figure 5: Qualitative Grad-CAM comparison (seed 0). Columns are the three fault classes (IR, OR, B); the top row is the random-window model (R), the bottom row the recording-level model (S). Shaded green marks the true fault band B, orange the 120 Hz-shifted control band B'. The random-window saliency (red fill) concentrates inside the true band more sharply, yielding a larger per-window margin; the recording-level saliency is more diffuse. This is the visual signature of the ILG.

10 Cross-Dataset Transfer

A single dataset cannot establish that the ILG is a general phenomenon rather than a quirk of the drive-end sensor. We therefore repeat the entire pipeline on a second dataset: the CWRU 12 kHz *fan-end* (FE) accelerometer. The fan-end sensor sits at a different location with a different transmission path and resonance behaviour, so it is a genuinely independent test of transferability while keeping the same fault physics and protocol machinery. The FE subset yields 49 recordings and 5864 windows; split overlaps are again verified ($R = 48$, $S = 0$).

Table 3 and Figure 6 report the two headline effects on both datasets. The generalisation gap and the ILG are *positive on both* sensors: leakage inflates accuracy and interpretability appearance regardless of sensor location. The magnitudes differ (FE shows a larger accuracy gap but a smaller ILG), which is expected given the weaker fault-band structure at the fan-end, but the sign and direction of the effect are stable. On the fan-end the ILG is also statistically significant over ten seeds (bootstrap 95% CI $[0.022, 0.073]$, paired-t $p = 0.008$, positive on nine of ten seeds).

Table 3: Cross-dataset transfer of leakage effects (ten-seed means; CI is a seed-level bootstrap 95% interval on the ILG).

Dataset	Acc. R	Acc. S	Gen. gap
DE (drive-end)	0.966 ± 0.013	0.913 ± 0.040	$+0.053$
FE (fan-end)	0.896 ± 0.021	0.835 ± 0.062	$+0.061$
Dataset	Margin R	Margin S	ILG (95% CI)
DE (drive-end)	0.263	0.168	$+0.095$ $[0.079, 0.110]$
FE (fan-end)	0.151	0.103	$+0.048$ $[0.022, 0.073]$

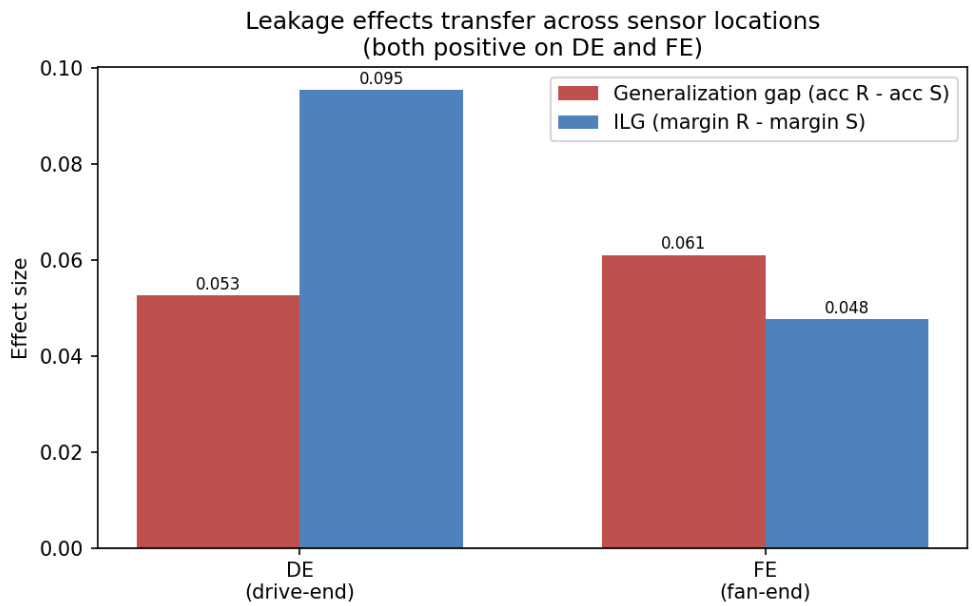

Figure 6: Generalisation gap and ILG on the drive-end and fan-end datasets. Both effects are positive on both sensors, indicating the leakage phenomena transfer across sensor location.

11 Ablations

Low-data corroboration. Restricting training to K windows per recording, the R$-$S accuracy gap persists and grows as the budget increases: on DE $+0.011$ (K=10), $+0.027$ (K=25), $+0.062$ (K=50); on FE $+0.058$, $+0.096$, $+0.070$. Thus the leakage-driven accuracy inflation is present across training budgets and is not an artefact of the full-data regime.

Grouping variable. Splitting protocol S by different units (ten seeds, DE) yields: by recording 0.948 ± 0.033, by load 0.940 ± 0.019, and by fault size 0.371 ± 0.225. The collapse to ~ 0.37 when testing on unseen fault sizes reveals that the model leans heavily on size-dependent features—a shortcut deeper than recording identity, and a caution against claiming size-invariant diagnosis. The same pattern holds on FE (by fault size 0.548 ± 0.156 versus by recording 0.821 ± 0.046).

Bandpass ablation. Removing the resonance band-pass sharply reduces the safe-split Grad-CAM margin: on DE it falls from 0.168 to 0.010, and on FE it becomes *negative* (-0.033), all over ten seeds. This shows the resonance band-pass is essential to physical consistency—the low-frequency envelope of the raw signal alone does not localise fault bands—and confirms that the representation choice is what makes the physical-consistency measure meaningful.

What the ablations jointly establish. Read together, the three ablations discharge three distinct alternative explanations. The low-data ablation rules out "the effect only exists because full-data accuracy saturates": the accuracy gap is stable across training budgets, so leakage inflation is not merely a saturation artefact. The grouping-variable ablation rules out "recording identity is the only shortcut": fault size is an even stronger shortcut (accuracy collapses to ~ 0.40 on unseen sizes), which both bounds the generality of any size-invariance claim and shows that our recording-level protocol is a conservative, not an adversarial, choice of split. The band-pass ablation rules out "the margin measures an arbitrary property of the network": without the resonance band-pass the margin nearly vanishes (DE) or reverses (FE), so the margin is genuinely tied to the physical representation and not to incidental network structure. Each ablation therefore removes a specific way the headline claim could have been an accident.

12 Discussion

Accuracy and interpretability respond to leakage on different scales. On a saturated benchmark, accuracy barely moves, so a practitioner who compares only accuracy may conclude that leakage is benign. The Grad-CAM margin exposes what accuracy hides: the random-window model's explanations are sharpened by memorisation rather than by better physics. This concretely supports the course principle of validating mechanism before trusting a metric. It also suggests a practical reporting recommendation: alongside a recording-level accuracy, report an explanation-consistency margin *with* a negative control, and treat a large protocol-dependent margin as a leakage warning.

13 Broader Implications

Our narrow finding—that leakage inflates the appearance of a physically grounded model more visibly than it inflates accuracy—points to a broader tension in machine-learning evaluation. As benchmarks saturate, headline metrics lose their power to discriminate between models that are right for the right reasons and models that are right by memorisation. When every method scores above 0.99, accuracy stops being an experiment and becomes a ceiling. The field's instinct in that situation is to reach for a harder dataset; our results suggest a complementary move, namely to reach for a *different axis of measurement* on the same dataset. Explanation consistency is one such axis, but the underlying principle is more general: an evaluation is only as trustworthy as its ability to be *wrong*, and a negative control is what supplies that ability.

This reframes interpretability from a post-hoc, presentational concern into a diagnostic instrument. A saliency map that merely "looks reasonable" is weak evidence, because looking reasonable is exactly what a memorising model can counterfeit. What carries evidential weight is the *contrast* between a true physical hypothesis and a matched false one—here, the fault band versus a shifted band. The moment an explanation is required to distinguish signal from a plausible decoy, it becomes falsifiable, and falsifiability is what separates a measurement from a decoration. We would argue that any interpretability claim intended to build trust should ship with such a control by default.

Finally, the leakage we study is a small, well-behaved instance of a pattern that recurs wherever data are not independent: adjacent windows, repeated measurements of the same physical unit, patients with multiple visits, or successive frames of a video. In each case the unit of independence, not the unit of observation, is what a split must respect. The specific numbers here are tied to CWRU, but the discipline they demand—define the unit of independence, split on it, and validate the mechanism with a control—travels far beyond bearing diagnosis. Progress on saturated benchmarks may depend less on new architectures than on more honest instruments for deciding what a model has actually learned.

14 A Practical Leakage-Audit Protocol

The lesson of this study can be packaged as a short, reusable audit that a practitioner can run before trusting a vibration-diagnosis result. It generalises the specific steps we took into protocol-level advice, and each item maps to a concrete artefact in our repository.

Step 1: Declare the unit of independence. State explicitly what a single independent example is—a window, a recording, or a physical bearing—and justify it physically. On CWRU the honest unit is at least the recording, because adjacent windows of one recording have spectral cosine similarity 0.91 versus 0.49 across recordings. Declaring this unit is the single most consequential decision in the evaluation.

Step 2: Split on that unit and verify the overlap is zero. Grouping by the declared unit is necessary but not sufficient; the overlap must be checked numerically. We verify train–test recording overlap (R=62 leaky, S=0 safe) and treat any nonzero safe-split overlap as a bug, not a rounding detail.

Step 3: Report a metric that can move. On a saturated benchmark, accuracy cannot discriminate; a metric near its ceiling carries little information. Report at least one metric with headroom—here, the controlled Grad-CAM margin—so that leakage has somewhere to show up.

Step 4: Attach a negative control to every explanation. Never report a saliency statistic without a matched decoy. Our frequency-shifted band is the decoy; the margin between the true and shifted statistic is what carries evidential weight. This is the frequency-domain analogue of the sanity checks of [6].

Step 5: Ablate the shortcut, not just the accuracy. Probe whether a stronger shortcut than the one under study exists. Our fault-size grouping ablation revealed that size is a deeper shortcut than recording identity; without that probe we might have overclaimed size-invariant diagnosis.

Step 6: Show every seed, then test. Report per-seed numbers, not only means, so that near-separation is visible; then back it with a formal paired test and a bootstrap interval on enough seeds (we use ten) to make the claim quantitative rather than anecdotal. Transparent per-seed evidence and a significance test are complementary, not substitutes.

This six-step audit is deliberately architecture- and domain-independent; the CWRU numbers are merely one instantiation.

15 Threats to Validity

We organise the main risks to our conclusions using the standard construct/internal/ external taxonomy, and state how the design addresses each.

Construct validity: does the margin measure "physical consistency"? The margin could in principle be high for reasons unrelated to fault physics—for instance if the network fixated on any narrow band. Two design choices guard against this. The negative control equalises band width between the true and shifted bands, so a band-agnostic fixation would raise $\mathrm{PC}_{\mathrm{true}}$ and $\mathrm{PC}_{\mathrm{neg}}$ together and leave the margin near zero; and the band-pass ablation shows that removing the physical representation collapses the margin. The measured margins therefore reflect band-specific, physics-aligned attention rather than a generic sharpness of saliency.

Internal validity: is the ILG caused by the split, or by a confound? Because protocols R and S share representation, architecture, optimiser, epoch budget, batch size, and the ten seeds, the only manipulated variable is the unit of splitting. The difference-in-differences form of the ILG further cancels any protocol-independent bias of the representation. The chief residual confound is class-support imbalance between protocols (protocol S has larger, recording-aligned test folds); we report Macro-F1 and per-class recall alongside accuracy so that imbalance cannot silently drive the headline.

External validity: how far do the numbers generalise? The effect is demonstrated on two independent CWRU sensors (drive-end and fan-end) with consistent sign, which rules out a single-sensor artefact but not the shared laboratory origin of both. We therefore state the scope precisely: the *mechanism* (memorisation sharpening apparent focus) and the *recommendation* (report a controlled explanation-consistency margin under a recording-level split) are what we claim to generalise; the specific magnitudes are CWRU-specific and should be re-measured on any new machine. We make no field-readiness claim.

Statistical validity. We run ten seeds and report both a formal paired test and a seed-level bootstrap. The ILG is significant on both datasets (DE: 95% CI $[0.079, 0.110]$, paired-t $p = 1.1 \times 10^{-6}$; FE: 95% CI $[0.022, 0.073]$, $p = 0.008$), and the per-seed table (Table 2) shows the paired differences are positive on nine or ten of ten seeds. We treat the small full-data accuracy gap as corroborating, not primary, precisely because it is within seed variance.

16 Limitations

- CWRU is a laboratory benchmark with seeded faults; our results do not establish field readiness.
- Both datasets are from CWRU (drive-end and fan-end); transfer to a physically distinct machine (e.g. a gearbox) is untested, and only one representation (envelope spectrum) and one architecture are studied.
- Fault frequencies use nominal rpm; the shifted-band control mitigates but does not eliminate band-definition error.
- Normal has only three usable recordings, so its recording-level evaluation is high-variance.

- The full-data accuracy gap is small; the primary effect is on interpretability, not accuracy.

17 Conclusion

On CWRU four-class diagnosis, random-window leakage inflates accuracy by only ~ 5.3 points but inflates the apparent physical consistency of Grad-CAM explanations by ILG = +0.095 (95% CI $[0.079, 0.110]$, paired-t $p = 1.1 \times 10^{-6}$ over ten seeds), an effect we trace to memorisation of seen recordings. Both effects reproduce on an independent fan-end sensor (ILG = +0.048, 95% CI $[0.022, 0.073]$, gap +0.061), indicating the phenomenon is not an artifact of a single sensor location. Mechanism validation, not accuracy, is the reliable guard against leakage on saturated benchmarks. We recommend reporting recording-level splits together with an explanation-consistency margin and a negative control.

18 Future Work

Four extensions follow directly from the present design.

From recording-level to bearing-level splits. Our protocol S isolates recordings, but a still stricter unit exists: the physical bearing, in the sense of Hendriks et al. [2]. Because CWRU couples fault size with distinct physical bearings, a bearing-independent split is expected to lower the honest margin further and may widen the ILG. Re-running the ILG procedure under a bearing-wise split would test whether apparent interpretability degrades monotonically as the split unit approaches true independence—a natural "dose-response" version of our experiment.

Beyond a single representation and architecture. We fix the envelope-spectrum 1D-CNN to keep the split as the sole variable. The ILG is representation-agnostic in definition, so it can be recomputed for raw-waveform CNNs, spectrogram 2D-CNNs, and transformer encoders. A representation$\times$protocol grid would show whether the leakage-inflation of interpretability is universal or specific to frequency-domain models.

From protocol-level to per-recording statistics. Our ten-seed design already yields a significant paired test and a bootstrap confidence interval on the ILG. A finer analysis would model the margin at the level of individual recordings—for example a mixed-effects model with recording as a random effect—to separate between-recording variance from protocol effect and to identify which recordings contribute most to the memorisation gap.

Other explanation methods and other domains. The negative-control construction is not tied to Grad-CAM; Integrated Gradients or attention roll-out could be substituted, and the sanity-check philosophy of Adebayo et al. [6] suggests reporting several. More broadly, the same controlled-margin idea applies wherever explanations are checked against a domain prior with a matched decoy—electrocardiogram lead localisation, or seismological phase picking, for example.

Reproducibility Statement

Seeds $\{0, 1, \ldots, 9\}$ are fixed; normalisation uses training statistics only; split overlaps are verified (R=62, S=0). Commands are in `README.md`; the environment is pinned in `environment/requirements.txt` (`torch 2.9.1+cpu`, `scikit-learn 1.8.0`, `numpy 2.4.0`, `scipy 1.16.3`). Every number traces to `results/tables/*.json` and `results/logs/run_log.txt`.

AI-Use Disclosure

An AI coding assistant was used to scaffold the repository, adapt course experiment code (ch39/ch41/ch53) to real `.mat` data, implement the envelope/Grad-CAM/ILG pipeline, and draft this manuscript. All experiments were executed locally; every reported number was produced by the committed scripts and verified against saved artifacts. The assistant did not invent results or citations; all references were verified against their primary sources, and any missing evidence would be marked [MATERIAL GAP]. The human researcher is responsible for the scientific claims, leakage design, and interpretation. A M1–M7 integrity self-audit is in `ars/integrity_report.md`.

A Notation

Table 4 collects the symbols used throughout.

Table 4: Notation.

Symbol	Meaning
$x[n]$	sampled vibration window ($N = 2048$, $f_s = 12\,\text{kHz}$)
x_b, e	band-pass filtered signal; amplitude envelope
$E[m]$	envelope-spectrum magnitude (86 bins, 0–500 Hz)
f_r	shaft rotation frequency rpm/60
BPFO/BPFI/BSF	outer/inner-race, ball-spin fault frequencies
B, B'	true fault bands; 120 Hz-shifted control bands
c_f	normalised Grad-CAM saliency over frequency bins
$\text{PC}_{\text{true}}, \text{PC}_{\text{neg}}$	saliency fraction in B, in B'
margin	$\text{PC}_{\text{true}} - \text{PC}_{\text{neg}}$
ILG	$\text{margin}(R) - \text{margin}(S)$
R, S	random-window, recording-level protocols

B Dataset Composition

Table 5 lists the recording counts per class for both sensors, taken from the audited manifests (`data_manifest/manifest.csv` and `FE_manifest.csv`). On the drive-end sensor two files are unreadable and are dropped, so the usable count is 62 (Normal falls from four to three). The class imbalance at the *recording* level—only three or four Normal recordings—is the reason the recording-level protocol has high variance on the Normal-adjacent decisions and the reason we report class-weighted loss and Macro-F1 rather than accuracy alone.

Table 5: Recordings per class (before dropping unreadable files).

Sensor	Normal	IR	OR	B	Total
DE (drive-end)	4	16	28	16	64
FE (fan-end)	4	12	21	12	49

After windowing (2048-point windows, hop 1024, capped at 200 per recording), the drive-end set yields 7533 windows and the fan-end set 5864 windows. Note that the recording-level imbalance is milder at the window level, because the few Normal recordings are long; this is precisely why leakage matters, as a handful of long Normal recordings can dominate a random-window split.

C Hyperparameters and Configuration

All values are fixed across protocols, seeds, and datasets unless noted.

Table 6: Fixed experimental configuration.

Item	Value
Window length / hop	2048 / 1024 (50% overlap)
Max windows per recording	200
Resonance band-pass	2–4 kHz, 4th-order Butterworth
Envelope FFT range	0–500 Hz (86 bins, 5.86 Hz/bin)
CNN channels / kernels	16–32–64 / 5, 5, 3
Pooling / head	global average pool + linear
Optimiser	Adam, lr 10^{-3}
Epochs / batch size	20 / 64
Loss	class-weighted cross-entropy
Fault-band half-width	$\pm$15 Hz, harmonics 1–3
Control shift	+120 Hz
Seeds	$\{0, 1, \ldots, 9\}$ (ten)
Compute	CPU (`torch 2.9.1+cpu`)

D Per-Class Recall on the Recording-Level Protocol

Table 7 reports per-class recall for protocol S (drive-end) across all ten seeds, extracted from `results/tables/main_results.json`. The variance is concentrated in the IR and OR classes (IR recall ranges 0.75–1.00, OR 0.65–1.00); Normal is trivially separated in every seed. This is consistent with the aggregate observation that recording-level evaluation is harder and noisier than random-window evaluation.

Table 7: Per-class recall, protocol S, drive-end, by seed (ten seeds).

Seed	Normal	IR	OR	B
0	1.00	0.830	0.976	0.983
1	1.00	1.000	0.898	1.000
2	1.00	0.920	0.654	0.994
3	1.00	0.847	0.876	0.997
4	1.00	0.749	0.973	0.816
5	1.00	1.000	0.870	0.918
6	1.00	0.974	1.000	0.926
7	1.00	0.934	0.803	0.989
8	1.00	0.762	0.862	0.980
9	1.00	0.937	0.859	0.949

E Worked Example of the ILG Computation

To make the metric fully concrete we trace it through real seed-0 drive-end numbers from `main_results.json`. For the random-window model the saliency fractions are $\mathrm{PC}_{\mathrm{true}} = 0.602$ and $\mathrm{PC}_{\mathrm{neg}} = 0.326$, so the within-protocol margin is

$$\mathrm{margin}(R, \mathrm{seed}\ 0) = 0.602 - 0.326 = 0.276. \tag{9}$$

For the recording-level model on the same seed, $\mathrm{PC}_{\mathrm{true}} = 0.625$ and $\mathrm{PC}_{\mathrm{neg}} = 0.460$, giving

$$\mathrm{margin}(S, \mathrm{seed}\ 0) = 0.625 - 0.460 = 0.164. \tag{10}$$

Notice that $\mathrm{PC}_{\mathrm{true}}$ is comparable between the two models (0.602 vs 0.625); the protocol effect is carried mainly by the control term $\mathrm{PC}_{\mathrm{neg}}$, which is 0.134 higher under the safe split. Averaging each margin over the ten seeds (Table 2) gives $\overline{\mathrm{margin}}(R) = 0.263$ and $\overline{\mathrm{margin}}(S) = 0.168$, hence

$$\mathrm{ILG} = 0.263 - 0.168 = +0.095. \tag{11}$$

The difference-in-differences structure is visible here: the inner subtraction cancels the common $\mathrm{PC}_{\mathrm{true}} \approx 0.61$ level, and the outer subtraction reports what is left once the control has removed band-placement bias. A reader who reported only $\mathrm{PC}_{\mathrm{true}}$ would have seen a 0.020 difference and concluded "no effect"; the controlled margin reveals a 0.095 effect of the opposite practical significance.

F Reproduction Checklist

The full pipeline is reproduced by the following ordered commands (drive-end; replace with `--dataset FE` for the fan-end run), each writing artifacts consumed by the next:

1. `python src/make_manifest.py` — build the recording-level manifest.
2. `python src/preprocess.py` — cut windows, save `windows.npz`.
3. `python src/make_splits.py` — emit protocol R/S index sets; verify overlaps.

4. `python src/run_experiments.py` — train, evaluate, compute margins and ILG.
5. `python src/run_ablations.py` — low-data, grouping, and band-pass ablations.
6. `python src/mech_intra_record.py` — intra- versus cross-record similarity.
7. `python src/make_figures.py` & `make_cross_figures.py` — regenerate figures.

Determinism is enforced by fixed seeds and train-only normalisation statistics; every reported number traces to a file under `results/tables/` and a line in `results/logs/run_log.txt`.

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